

TONG WEI

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EDUCATION

Czech Technical University in Prague (CVUT)
Computer Science PhD candidate
Robust Estimation; Computer Vision; Deep Learning.

Sep. 2022 – present, Czech Republic
Supervisor: Jiří Matas, Daniel Barath

Eötvös Loránd University (ELTE)
Computer Science MSc. (AI Specialisation)
AI Principles; Methods and Tools for AI Applications; 3D Computer Vision; AI Robotics; Machine Learning.

Sep. 2020 – Jun. 2022, Hungary
Score: 4.36/5, Excellent graduate (5/5)

Liaoning Technical University (LNTU)
Communication Engineering BSc.
C++ Programming; Advanced Mathematics; Probability Theory and Statistics; Information Theory.

Aug. 2015 – Jun. 2019, China
Score: 3.19/4.5

PUBLICATIONS

- [1] **T. Wei, G. Tolias, J. Matas, and D. Barath. Global-Aware Edge Prioritization for Pose Graph Initialization.** *[CVPR'26 oral, award candidate][code]*
- [2] **T. Wei, P. Lindenberger, J. Matas, and D. Barath. Breaking the Frame: Visual Place Recognition by Overlap Prediction.** *[WACV'24][code]*
- [3] **T. Wei, Y. Patel, A. Shekhovtsov, J. Matas, and D. Barath, Generalized Differentiable RANSAC.** *[ICCV'23][code]*
- [4] **T. Wei, J. Matas, and D. Barath, Adaptive Reordering Sampler with Neurally Guided MAGSAC.** *[ICCV'23][code]*
- [5] Z. Tao, **T. Wei, J. Li, Wavelet Multi-level Attention Capsule Network for Texture Classification.** *IEEE Signal Processing Letters, 2021, 28: 1215-1219. [SPL'21]*
- [6] S. Lin, K. Chi, **T. Wei, and Z. Tao, Underwater Image Sharpening Based on Structure Restoration and Texture Enhancement.** *Applied Optics, 2021, 60 (15): 4443-4454. [Optica'21]*

PROJECTS & VISITING

Academic Guest at CVG, ETH Zurich, Oct. to Dec.
Minimal spanning tree-based image pair retrieval for Structure from Motion.

ETH Zurich, 2024

International Computer Vision Summer School (ICVSS)

Italy, 2023

Semester Project: Habit Tracker in Software Technology

ELTE, 2021

Collaborated on developing a habit tracking system with personalized recommendations and motivational quotes for diverse users. Implemented back-end services using FastAPI (Python).

National Project: Research on Common Key Technologies of Testing and Evaluation Services

LNTU, 2020

Focusing on point cloud classification algorithms using attention mechanisms and point projection-based features.

Bachelor Intern Project: 3D Object Detection for Self-Driving Vehicles

China, 2019

Working on learning-based 3D object detection algorithms using multiple sensors at Machine Vision and Pattern Recognition Group, Haixi Institutes, Chinese Academy of Sciences.

INVITED TALKS & SUPERVISION

Supervisor of a semester project at Mixed Reality course, ETH Zurich <i>Where am I? Interactive indoor localization and navigation on Magic Leap 2 hardware.</i>	Switzerland, 2024
Invited talk at Computer Vision Winter Workshop (CVWW) 2024 <i>Learning-based RANSAC scoring functions</i>	Slovenia, 2024
Invited talk at Computer Vision Winter Workshop (CVWW) 2023 <i>Differentiable RANSAC components and applications</i>	Austria, 2023

AWARD AND SCHOLARSHIP

Recipient of the Stipendium Hungaricum Master's Scholarship.	2020–2022
Honored as an Excellent Master's Student, the first semester, LNTU.	2021
First Prize (Band A), National English Competition for College Students (NECCS).	2020
Best Bachelor Thesis Award “3D Object Detection Network for Point Cloud Classification”, LNTU.	2019

OTHERS

Review Experience:	CVPR, ECCV, NeurIPS, ICLR
Technical Skills:	Computer Vision, 3D Reconstruction, Structure from Motion, Pose Graph Construction, Robust Estimation, Image Processing, Image Retrieval, Feature Matching, Deep Learning, Representation Learning
Programming Skills:	Python, C/C++, PyTorch, Keras, Korina, OpenCV
Language Skills:	English (fluent), Chinese (native)